Band-1 Computational Experiment: Legendre's Conjecture — Finite Square-Interval Verification

Human + ZoraASI full-dash (executor)

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Band-1 Computational Experiment: Legendre's Conjecture (Finite Verification)

Honesty stamp. This note **verifies** that for every integer n with $1 \le n \le N$ the open interval $(n^2, (n+1)^2)$ contains at least one prime, for an explicit finite N computed on this box. It does **not** prove **Legendre's conjecture** for all n. Finite-range verification $\neq$ conjecture. **Status: OPEN.** This is **not** a Clay Millennium claim and **not** a ToE / GATE-04 unlock. Do **not** attribute a proof to Legendre himself for this statement as a theorem for all n.

Abstract

We scan each open square interval $(n^2, (n+1)^2)$ for $n = 1, \dots, N$ using deterministic-strength primality tests (`gmpy2.is_prime`, Baillie–PSW) on odd candidates and obtain

$$\mathbf{N = 10{,}000{,}000 = 10^7}, \qquad \mathbf{failures = 0}.$$

Wall time for the primary run: **60.18 s** (total wall including unit tests and intermediate $N = 10^6$: **65.25 s**). An intermediate certificate at $N = 10^6$ also recorded **0** failures (5.07 s). This is a Band-1 **range-collapse / methods** certificate far below the published computational record. **Conjecture status: OPEN.**

Problem statement

Legendre's conjecture (one of Landau's problems, 1912). For every integer $n \ge 1$ there exists a prime p with

$$n^2 < p < (n+1)^2.$$

This experiment. Prove computationally that the statement holds for all $n \in \{1, \dots, N\}$ for a chosen explicit N tractable on this box (minutes), and record the literature ledger of larger verified ranges (including the stronger Oppermann conjecture).

Literature ledger (not re-proved on box)

Year	Result	Authors	Status
1912	Listed among Landau's four problems	Landau (ICM)	historical framing
$ $2014	Legendre/Oppermann to $n \le 2 \cdot 10^9$ (via primes to $4 \cdot 10^{18}$)	via Oliveira e Silva–Herzog–Pardi tables	prior computational bound

Year	Result	Authors	Status
2024/2025	**Oppermann (hence Legendre) for all** $n \le 7.05 \cdot 10^{13}$	**Sorenson & Webster**, *Research in Number Theory* **11**	**published record**
2026	Conditional weakenings under RH for higher powers $2+\delta$	Chamberland–Straub (arXiv:2602.22502) + CMS	**conditional only** — not unconditional Legendre

Primary citation for the published verification record: J. Sorenson, J. Webster, *An algorithm and computation to verify Legendre's conjecture up to* $7 \cdot 10^{13}$, Research in Number Theory **11** (2025); published online 2024-12-06 (DOI 10.1007/s40993-024-00589-4). They verify **Oppermann's** conjecture (primes in $(n^2, n(n+1))$ and $(n(n+1), (n+1)^2)$), which strictly implies Legendre, up to $N = 7.05 \cdot 10^{13}$.

Unconditional Legendre for all n remains OPEN, including under RH for consecutive squares ($\alpha = 2$); RH-conditional results address larger exponents $2+\delta$ (see ledger).

Method (box)

1. **Unit tests.** Check $n = 1, 2, 10$; full verify $n = 1..30$; sanity on `gmpy2.is_prime`.
2. **Interval scan.** For each $n = 1..N$, test odd integers in $(n^2, (n+1)^2)$ with `gmpy2.is_prime` (Baillie–PSW) until the first prime is found (or the interval ends).
3. **Failure log.** Record any n with no prime in the open interval (expect **none**).
4. **Diagnostics.** Track the offset $p - n^2$ of the first prime found; report max/mean.

Complexity (heuristic). Expected prime gap near n^2 is $\sim 2 \ln n$, so expected work is near-linear in N times a slow-growing factor — matching the observed ~ 60 s at $N = 10^7$.

Not used as a claim surface: full sieves to $(N+1)^2$ (unnecessary for existence), Oppermann's stronger split intervals (literature only), RH-conditional theorems.

Results (numbers!)

Unit tests

`PASS`: primes in $(1, 4)$, $(4, 9)$, $(100, 121)$; full verify $n = 1..30$ with **0** failures; `gmpy2` prime/composite sanity.

Box verification

Quantity	Value
Intermediate N	10^6
Failures @ 10^6	**0**
Wall @ 10^6	5.0659 s
Primary N verified	$10{,}000{,}000 = 10^7$
Integers n checked	10,000,000
Failures	**0**
All verified	**True**
Max first-prime offset $p - n^2$	**541** at $n = 4{,}292{,}936$
Mean first-prime offset	**27.8708**
Primary wall	**60.1822 s**
Total wall (tests + both N)	**65.2514 s**

Sample first primes found in $(n^2, (n+1)^2)$

n	n^2	first prime found
1	1	3
2	4	5
10	100	101
100	10{,}000	10{,}007
1{,}000	10^6	1{,}000{,}003
10{,}000	10^8	100{,}000{,}007
100{,}000	10^{10}	10{,}000{,}000{,}019
1{,}000{,}000	10^{12}	1{,}000{,}000{,}000{,}039
10{,}000{,}000	10^{14}	100{,}000{,}000{,}000{,}031

What this does not claim

- Legendre's conjecture for all n (still **OPEN**).
- A proof by Legendre, or resolution of Landau's problem.
- Oppermann's / Andrica's / Brocard's conjectures.
- Clay Millennium / ToE / GATE-04 unlock.
- Matching or exceeding the Sorenson–Webster $7.05 \cdot 10^{13}$ record on this box.

Reproducibility

```
python3 experiments/band1/legendre/legendre_verify.py
# writes results.json, literature_ledger.json, SUMMARY.txt
```

Requires: Python 3, `gmpy2`.

Conclusion

For all $n = 1, \ldots, 10^7$, a prime exists in $(n^2, (n+1)^2)$ (**0** failures; wall ~ 60 s). Literature verifies much farther ($n \leq 7.05 \cdot 10^{13}$, Oppermann). **Legendre's conjecture remains OPEN.**